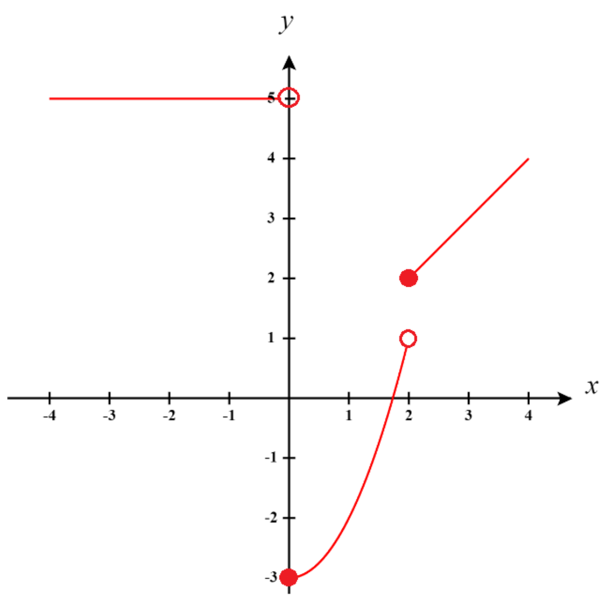
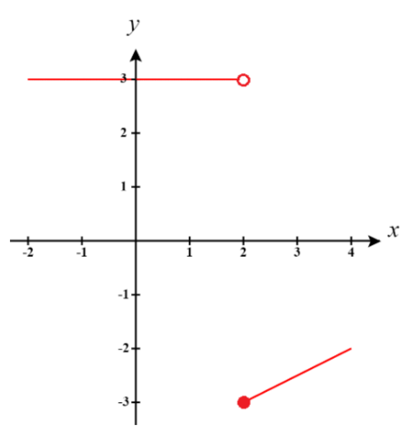
MTH 200 **Skills Check 1 Worksheet**

Skills checks cover pre-requisite material needed for the first unit of the course, limits. Answer the following questions neatly and completely. Calculators are not allowed.

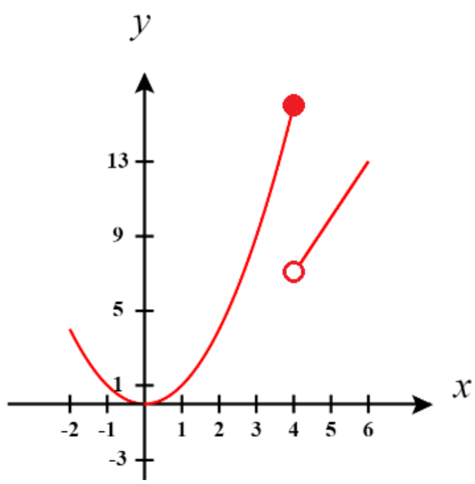
1. Graph the following piecewise function: $f(x)=\begin{cases} x & if \quad x\geq 2 \\ x^2-3 & if \quad 0\leq x<2 \\ 5 & if \quad x<0 \end{cases}$



2. Graph the following piecewise function: $f(x)=\begin{cases} \frac{1}{2}x-4 & if \quad x\geq 2 \\ 3 & if \quad x<2 \end{cases}$



3. Graph the following piecewise function: $f(x)=\begin{cases} 3x-5 & if \quad x>4 \\ x^2 & if \quad x\le 4 \end{cases}$



4. Complete the following subtraction and fully simplify the result: $\dfrac{3}{t+3}-\dfrac{1}{4t+12}$

$$\frac{3}{t+3}-\frac{1}{4t+12}=\frac{3}{t+3}-\frac{1}{4(t+3)}=\frac{3}{t+3}\cdot\frac{4}{4}-\frac{1}{4(t+3)}=\frac{12}{4(t+3)}-\frac{1}{4(t+3)}=\frac{11}{4(t-3)}$$

5. Complete the following addition and fully simplify the result: $\dfrac{x}{2x^2+2x}+\dfrac{3}{x+1}$

$$\frac{x}{2x^2+2x}+\frac{3}{x+1}=\frac{x}{2x(x+1)}+\frac{3}{x+1}=\frac{x}{2x(x+1)}+\frac{3}{x+1}\cdot\frac{2x}{2x}=\frac{x}{2x(x+1)}+\frac{6x}{2x(x+1)}=\frac{7x}{2x(x+1)}=\frac{7}{2(x+1)}$$

6. Complete the following subtraction and fully simplify the result: $\dfrac{5}{x-2}-\dfrac{4x+13}{x^2+x-6}$

$$\frac{5}{x-2}-\frac{4x+13}{x^2+x-6}=\frac{5}{x-2}-\frac{4x+13}{(x+3)(x-2)}=\frac{5(x+3)}{(x-2)(x+3)}-\frac{4x+13}{(x+3)(x-2)}=\frac{5x+15-4x-13}{(x-2)(x+3)}=\frac{x+2}{(x-2)(x+3)}$$

7. Determine the interval communicated by the inequality: $|x+3|<5$

In order for the given statement to be true we know that:

$$-5<x+3<5 \quad \Leftrightarrow \quad -8<x<2$$

8. Determine the interval communicated by the inequality: $3|2x-1|<9$

$3|2x-1|<9 \;\rightarrow\; |2x-1|<3$

In order for the given statement to be true we know that:

$$-3<2x-1<3 \quad \Leftrightarrow \quad -2<2x<4 \quad \Leftrightarrow \quad -1<x<2$$

9. Determine the interval communicated by the inequality: $|1-2x|<5$

In order for the given statement to be true we know that:

$$-5<1-2x<5 \quad \Leftrightarrow \quad -6<-2x<4 \quad \Leftrightarrow \quad -2<x<3$$

10. Rationalize the denominator of the fraction: $\dfrac{x-4}{\sqrt{x}-2}$

$$\frac{x-4}{\sqrt{x}-2}=\frac{x-4}{\sqrt{x}-2}\cdot\frac{\sqrt{x}+2}{\sqrt{x}+2}=\frac{(x-4)\left(\sqrt{x}+2\right)}{x-4}=\sqrt{x}+2$$

11. Rationalize the numerator of the fraction: $\dfrac{\sqrt{t^2+9}-3}{t^2}$

$$\frac{\sqrt{t^2+9}-3}{t^2}=\frac{\sqrt{t^2+9}-3}{t^2}\cdot\frac{\sqrt{t^2+9}+3}{\sqrt{t^2+9}+3}=\frac{\left(t^2+9\right)-9}{t^2\left[\sqrt{t^2+9}+3\right]}=\frac{1}{\sqrt{t^2+9}+3}$$

12. Rationalize the denominator of the fraction: $\dfrac{w-2}{\sqrt{w+2}-2}$

$$\frac{w-2}{\sqrt{w+2}-2}=\frac{w-2}{\sqrt{w+2}-2}\cdot\frac{\sqrt{w+2}+2}{\sqrt{w+2}+2}=\frac{(w-2)\left[\sqrt{w+2}+2\right]}{(w+2)-4}=\sqrt{w+2}+2$$

13. Setup and simplify the expression $\dfrac{f(x+h)-f(x)}{h}$ using the function: $f(x)=5x^2-1$

$$\frac{\left[5(x+h)^2-1\right]-\left[5x^2-1\right]}{h}=\frac{5x^2+10xh+5h^2-1-5x^2+1}{h}=\frac{10xh+5h^2}{h}=10x+5h$$

14. Setup and simplify the expression $\dfrac{f(x+h)-f(x)}{h}$ using the function: $f(x)=4-2x^2$

$$\frac{\left[4-2(x+h)^2\right]-\left[4-2x^2\right]}{h}=\frac{4-2\left(x^2+2xh+h^2\right)-4+2x^2}{h}=\frac{-4xh-2h^2}{h}=-4x-2h$$

15. Setup and simplify the expression $\dfrac{f(x+h)-f(x)}{h}$ using the function: $f(x)=3x^2+1$

$$\frac{\left[3(x+h)^2+1\right]-\left[3x^2+1\right]}{h}=\frac{3\left(x^2+2xh+h^2\right)+1-3x^2-1}{h}=\frac{6xh+3h^2}{h}=6x+3h$$